

SF2940 Probability Theory – Lecture plan

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Information

This pdf (which will be updated throughout the course) outlines the lecture plans for the course SF2940.

When referring to the course literature in this document, we write [K] for *Lecture Notes: Probability and Random Processes at KTH* by Timo Koski and [G] for *An Intermediate Course in Probability* by Allan Gut.

I recommend reading Gut's book, as I consider that the superior material. Reading instructions for Koski's notes is included for those that like that one.

Week 1

This week will cover probability spaces, distributions, random variables, multivariate random variables and functions of random variables.

Lecture 1

Probability spaces, distributions, random variables.

[K]: Sections 1.1-1.6 and 1.8.

[G]: Introduction: Sections 2-8.

Lecture 2

Multivariate random variables, functions of random variables.

[G]: Chapter 1 – Sections 1.1-1.2, pages 15-24.

[K]: Chapter 2 – Sections 2.1-2.4.

Week 2

Lectures 3 and 4

Conditional distributions, conditional expectation, distributions with random parameters.

[G]: Chapter 2 – Sections 2.1-2.3.

[K]: Chapter 3 – Sections 3.1-3.6.

Week 3

Lectures 5 and 6

Generating functions (probability generating function, moment generating function, characteristic function) and their applications to sums of random variables – including sums of random numbers of random variables.

[G]: Chapter 3 – Sections 3.1-3.6.

[K]: Chapters 4 and 5 – Sections 4.1-4.5 and 5.1-5.5, 5.7.

Week 4

Lecture 7

Convergence of random variables. Uniqueness of limits.

[G]: Chapter 6 – Sections 6.1-6.2.

[K]: Chapter 6 – Sections 6.1-6.2.

Lecture 8

Hierarchy/relationships between modes of convergence.

[G]: Chapter 6 – Section 6.3.

[K]: Chapter 6 – Sections 6.3.

Week 5

Lecture 9

Convergence via transformations, the law of large numbers, the central limit theorem, Cramér-Slutsky's theorem.

[G]: Chapter 6 – Sections 6.4-6.6.

[K]: Chapter 6 – Sections 6.4-6.7.

Lecture 10

Multivariate normal distribution: Linear algebra preliminaries, covariance matrices, three definitions of multivariate normal distribution.

[G]: Chapter 5 – Sections 5.1-5.5.

Week 6

Lecture 11

Multivariate normal distribution: conditional distributions, independence, linear transformations,

[G]: Chapter 5 – Sections 5.6-5.7.

Lecture 12

Branching processes.

[G]: Chapters 3 and 6 – Sections 3.7 and 6.7.

Week 7

Lecture 13

Branching processes and random walks.

[G]: Chapters 3 and 6 – Sections 3.7 and 6.7.

Lecture 14

Solving old exams: First exam of 2025 and perhaps part of the first exam of 2024.